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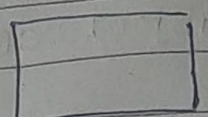
Static & Final in Java

Static

A variable which a property of class rather than object, we used static

```
class student {
    String name;
    int age;
    int rollno;
    static string college;
}
```

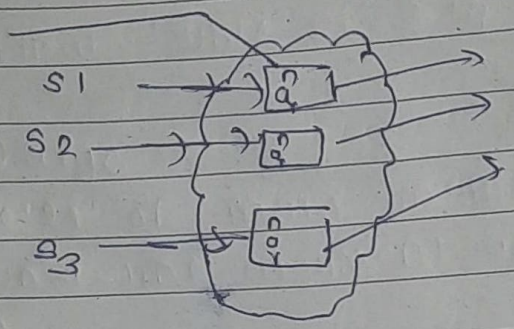
Instance variable (obj var)



developing application for students from same clg.

class variable

Inside obj, only obj variables are stored.



college = "ITTG"

not stored in heap

```
student s1 = new student();
s1.name = "Aditya";
s1.college = "ITTG";  ← also valid
student.college = "ITTG";
```

↓

```
classname.variable = value;
```

* Static keyword can be used with Method also.

```
student.MarkAttendance();
s1.MarkAttendance();
```

A static method is class method

So, static can be used on
- ↳ variable
- ↳ Method
- ↳ class (If nested)
- ↳ Param ✗
- ↳ Block ✓

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method → static

Rules of static

↳ ① one static method can only call other static Method.

② static Method can only access static variable

③ Does not have access to this keyword.

Static block

↳ used to initialize static variables

syntax

```
static {
        college = "ITT Guhawati";
}
```

- A Method parameters can not be static, because their are local variables, their scope ends as soon as func end
- A class cannot be static (on rule level). But if class is nested class then it can be static

Final Keyword

- Final key word is used to restrict Modification, essentially making an entity "Finalized".

Final means Fixed, constant.

- eg:

```
class Random {
        final double PI = 3.14;
}
```

Final keyword can be used ~~HA~~ with
↳ variable
↳ Method
↳ class
↳ Method parameter

Print (final string name)

* why main is static in Java

• static final double pi = 3.14

* String [] arge
↳ parameter of main method that stores command-line argument as an array of string.

Final conclusion

Static is a part of class and only exist within metadata of class and can be changed any number of time.

While final variable is a part of object & could be different for each object. if 1000 ob all obj will have their separate final variable but once assigned it can't be changed.